

COVID-19 Data Visualisation Analysis

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This document presents a comprehensive data visualisation analysis of global and European COVID-19 trends using both **Python-based visualisations** and a **Tableau dashboard**. Through a series of visualisations, it explores vaccination coverage, vaccine diversity, infection rates, and mortality comparisons across different regions and countries. The analysis integrates data from WHO COVID-19 global statistics and vaccination datasets, while the Tableau dashboard provides an interactive overview that enhances data exploration and comparison. Overall, the document demonstrates effective use of multiple visualisation tools to clearly communicate complex public health data in an accessible and visually engaging manner.

Python Visualisations: Page 2 - 22

Tableau dashboard: Page 23

References: Page 24

Visualisation 1

What is the percentage of the population that has been fully vaccinated in the whole world?

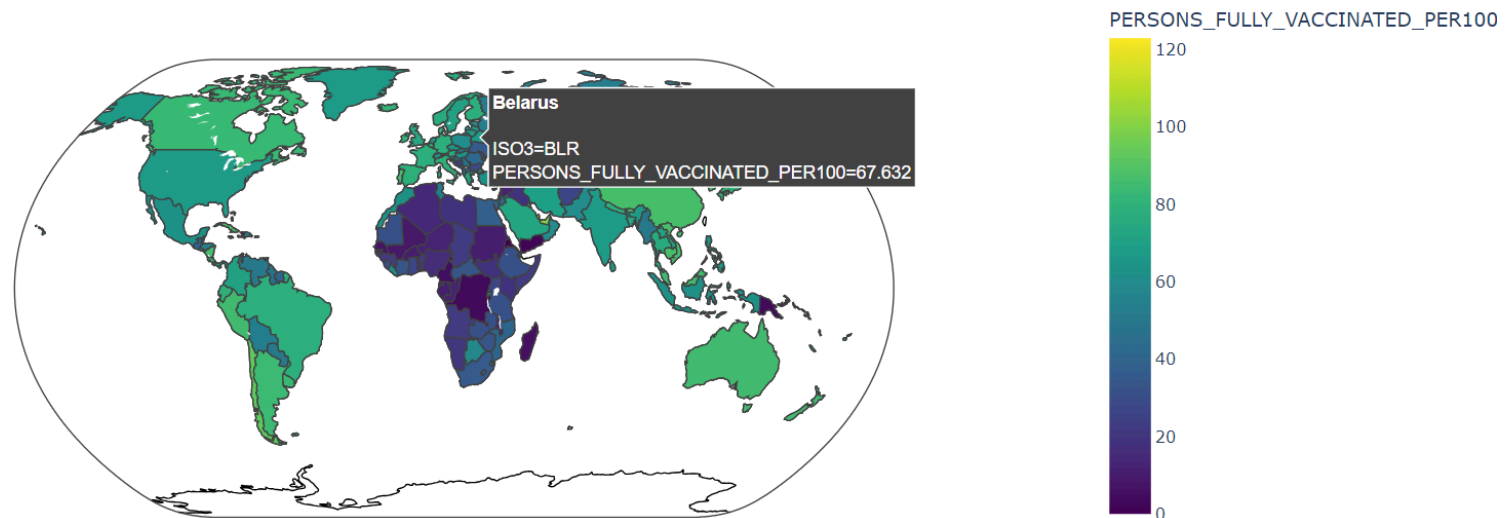
```
#1 What is the percentage of the population that has been fully vaccinated in different countries?
# Import necessary libraries
import plotly.express as px # Import Plotly Express for visualization
import pandas as pd # Import pandas for data handling

# Load vaccination data from a CSV file
vaccination_data = pd.read_csv('vaccination-data.csv')

# Create a choropleth map using Plotly Express
fig = px.choropleth(vaccination_data, # Use the vaccination data as the source
                    locations='ISO3', # Use ISO3 codes to uniquely identify countries
                    color='PERSONS_FULLY_VACCINATED_PER100', # Color countries based on fully vaccinated
                    percentage
                    hover_name='Country', # Display country names when hovering over them
                    color_continuous_scale='Viridis', # Choose a color scale for the map
                    projection='natural earth' # Use a natural earth projection for the map
                    )

# Update the layout of the figure
fig.update_layout(
    title='Fully Vaccinated Population Percentage by Country', # Set the title of the map
    geo=dict(
        showcoastlines=True, # Display coastlines on the map
        coastlinecolor="Black", # Set the color of the coastlines to black
        showland=True, # Display land areas on the map
        landcolor="white" # Set the color of land areas to white
    )
)

# Show the figure
fig.show() # Display the choropleth map
```



The choropleth map visualizes the global distribution of fully vaccinated populations by country. Darker colors indicate higher percentages, while lighter shades represent lower vaccination rates. The map employs a natural earth projection with clear coastlines in black and white land areas for easy interpretation.

The title "Fully Vaccinated Population Percentage by Country" concisely communicates the main theme, and hovering over a country displays its name for added context. In the screenshot above, I have hovered my mouse over Belarus. This visualisation effectively conveys vaccination trends across nations. The information is extracted from the "vaccination-data.csv" file only.

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Visualisation 2:

What are the top 15 countries in the world that have used the highest number of vaccine types?

```
# Import necessary libraries
import pandas as pd
import matplotlib.pyplot as plt

# Read the CSV file into a DataFrame
df = pd.read_csv('vaccination-data.csv')

# Sort the DataFrame by 'NUMBER_VACCINES_TYPES_USED' in descending order and select top 15
top_15_countries = df.nlargest(15, 'NUMBER_VACCINES_TYPES_USED')

# Define a list of colors (customize this list as needed)
colors = ['blue', 'green', 'red', 'purple', 'orange']

# Set the figure size
plt.figure(figsize=(12, 8))

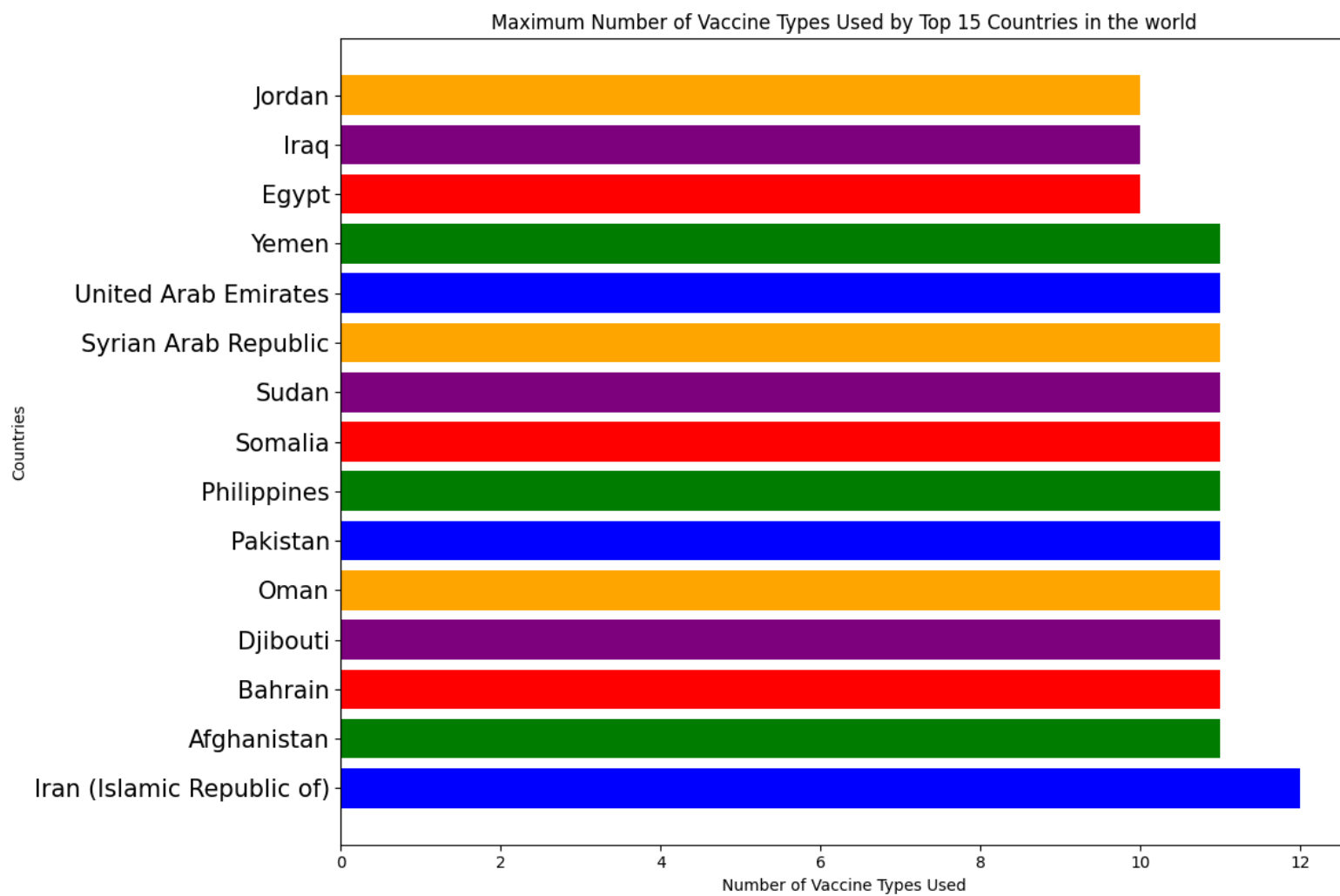
# Create a horizontal bar chart with different colors
plt.barh(top_15_countries['Country'], top_15_countries['NUMBER_VACCINES_TYPES_USED'], color=colors)

# Add labels and title
plt.xlabel('Number of Vaccine Types Used') # X-axis label
plt.ylabel('Countries') # Y-axis label
plt.title('Maximum Number of Vaccine Types Used by Top 15 Countries in the world') # Chart title

# Set custom y-ticks with larger font size for better readability
plt.yticks(range(len(top_15_countries['Country'])), top_15_countries['Country'], fontsize=15)

# Ensure all labels and ticks fit within the figure
plt.tight_layout()

plt.show()
```



The provided visualisation displays top 15 countries in the world that have used the highest number of vaccine types. Each horizontal bar represents a country, with the length of the bar corresponding to the number of vaccine types utilized. The chart employs a distinct color for each country, making it easy to differentiate between them.

The figure's y-axis denotes the countries, while the x-axis quantifies the number of vaccine types employed. This visualisation is particularly useful for understanding the varying approaches and strategies adopted by countries in managing their vaccination campaigns. The clear layout and color-coded bars enable viewers to discern trends and disparities at a glance. The information is extracted from “vaccination-data.csv” only.

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Visualisation 3:

What percentage of the population is fully vaccinated compared to the percentage of deaths in Northern and Western European Countries?

```
# Import necessary libraries
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.ticker import MultipleLocator

# Define a list of Northern and Western European countries
northern_western_european_countries = ['Sweden', 'Norway', 'Denmark', 'Finland', 'Iceland',
'Netherlands', 'Belgium', 'Luxembourg', 'France', 'Germany', 'United Kingdom', 'Ireland']

# Load data from the first CSV file
df_covid_data = pd.read_csv('WHO-COVID-19-global-table-data.csv')

# Load data from the second CSV file
df_vaccination_data = pd.read_csv('vaccination-data.csv')

# Filter data for Northern and Western European countries
df_covid_northern_western =
df_covid_data[df_covid_data['Country'].isin(northern_western_european_countries)]
df_vaccination_northern_western =
df_vaccination_data[df_vaccination_data['Country'].isin(northern_western_european_countries)]

# Calculate percentage of deaths per 100,000 population
df_covid_northern_western['Deaths_percentage'] = (df_covid_northern_western['Deaths - cumulative
total per 100000 population'] / 100000) * 100

# Calculate percentage of persons fully vaccinated
```

```

df_vaccination_northern_western['Fully_Vaccinated_percentage'] =
df_vaccination_northern_western['PERSONS_FULLY_VACCINATED_PER100']

# Merge DataFrames based on a common identifier (e.g., 'Country')
merged_df = pd.merge(df_covid_northern_western, df_vaccination_northern_western, left_on='Country',
right_on='Country', how='inner')

# Create a scatter plot with larger figure size
plt.figure(figsize=(12, 10))

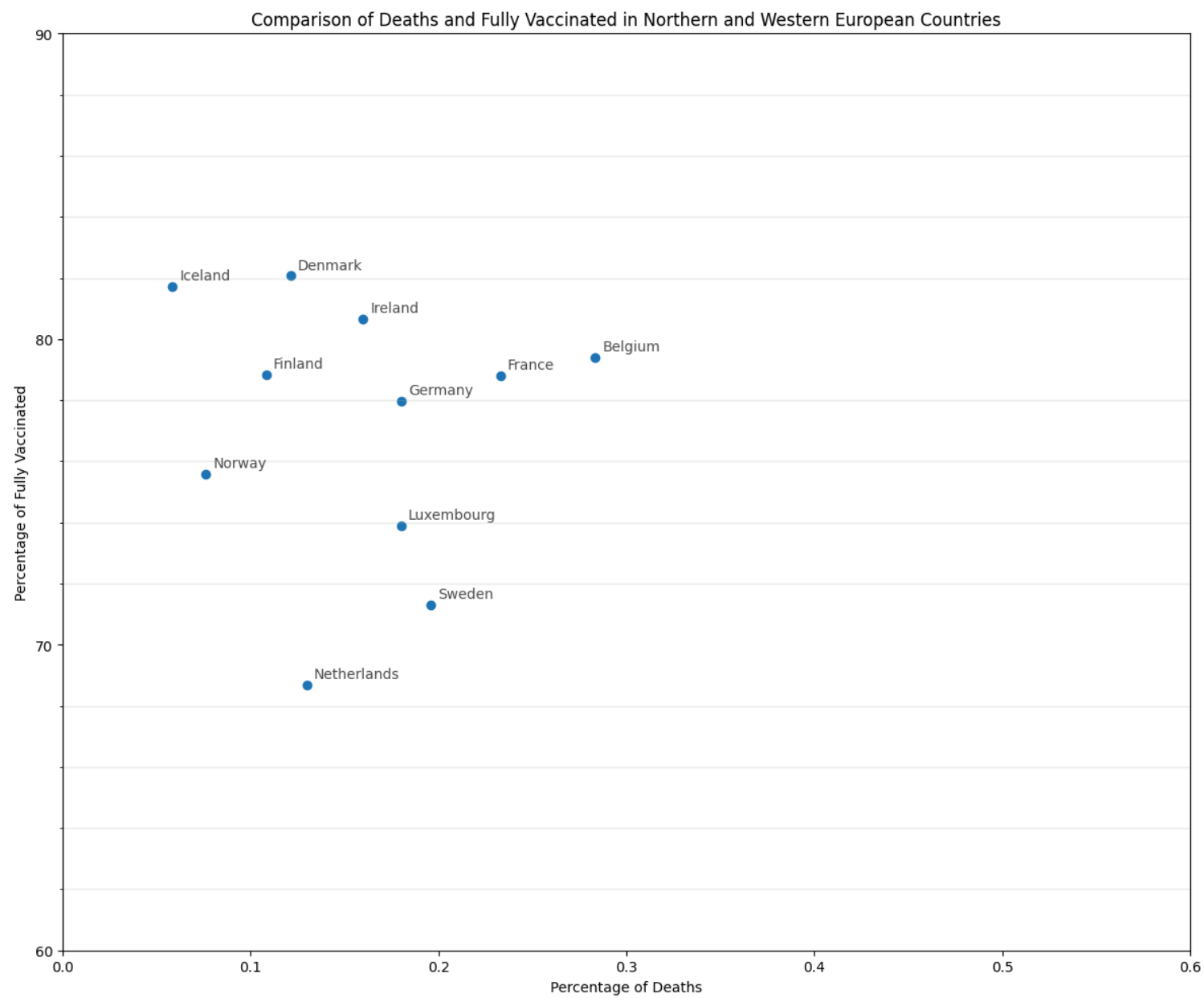
# Add scatter plot
plt.scatter(merged_df['Deaths_percentage'], merged_df['Fully_Vaccinated_percentage'])
plt.xlabel('Percentage of Deaths')
plt.ylabel('Percentage of Fully Vaccinated')
plt.title('Comparison of Deaths and Fully Vaccinated in Northern and Western European Countries')

# Annotate data points with country names and adjust positions
for i, country in enumerate(merged_df['Country']):
    plt.annotate(country, (merged_df['Deaths_percentage'][i],
merged_df['Fully_Vaccinated_percentage'][i]), fontsize=10, alpha=0.7, xytext=(5,5),
textcoords='offset points')

# Adjust the limits and ticks of the axes
plt.xlim(0, 0.6) # Set x-axis limits from 0 to 0.6
plt.ylim(60, 90) # Set y-axis limits from 60 to 90
plt.yticks(range(60, 91, 10)) # Set y-axis ticks to go up in 10s
plt.gca().yaxis.set_minor_locator(MultipleLocator(2)) # Increase the gap between minor ticks
plt.xticks([i/10 for i in range(0, 7)]) # Set x-axis ticks to go up in 0.1 increments
plt.grid(which='minor', alpha=0.2) # Add gridlines for minor ticks

# Show the plot
plt.tight_layout()
plt.show()

```

This visualisation combines data from two datasets: the "WHO-COVID-19-global-table-data" dataset, which provides information on COVID-19-related metrics, and the "vaccination-data" dataset, which offers data on vaccination rates. The code specifically selects Northern and Western European countries from the two datasets.

The scatter plot compares the percentage of COVID-19 related deaths (x-axis) and the percentage of fully vaccinated individuals (y-axis) for Northern and Western European countries. Each point on the plot represents a specific country, allowing for a quick visual comparison of these key metrics across the region.

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Visualisation 4:

What is the spread between the percentage of Covid Cases in European Countries?

```
# Import necessary libraries
import pandas as pd
import matplotlib.pyplot as plt

# Load data from the CSV file
df_covid_data = pd.read_csv('WHO-COVID-19-global-table-data.csv')

# Filter data for European countries
df_covid_europe = df_covid_data[df_covid_data['WHO Region'] == 'Europe']

# Calculate percentage of cases per 100,000 population
df_covid_europe['Cases_percentage'] = (df_covid_europe['Cases - cumulative total per 100000
population'] / 100000) * 100

# Set up figure and axis for the boxplot
plt.figure(figsize=(10, 8))
ax = plt.gca()

# Create boxplot for Cases percentages
boxplot_data = [df_covid_europe['Cases_percentage']] # Data to be plotted
positions = [1] # Position of the boxplot on the x-axis
labels = ['Cases %'] # Labels for the boxplot

# Create the boxplot
plt.boxplot(boxplot_data, positions=positions, labels=labels, widths=0.6, patch_artist=True,
vert=True, notch=True, showfliers=False)

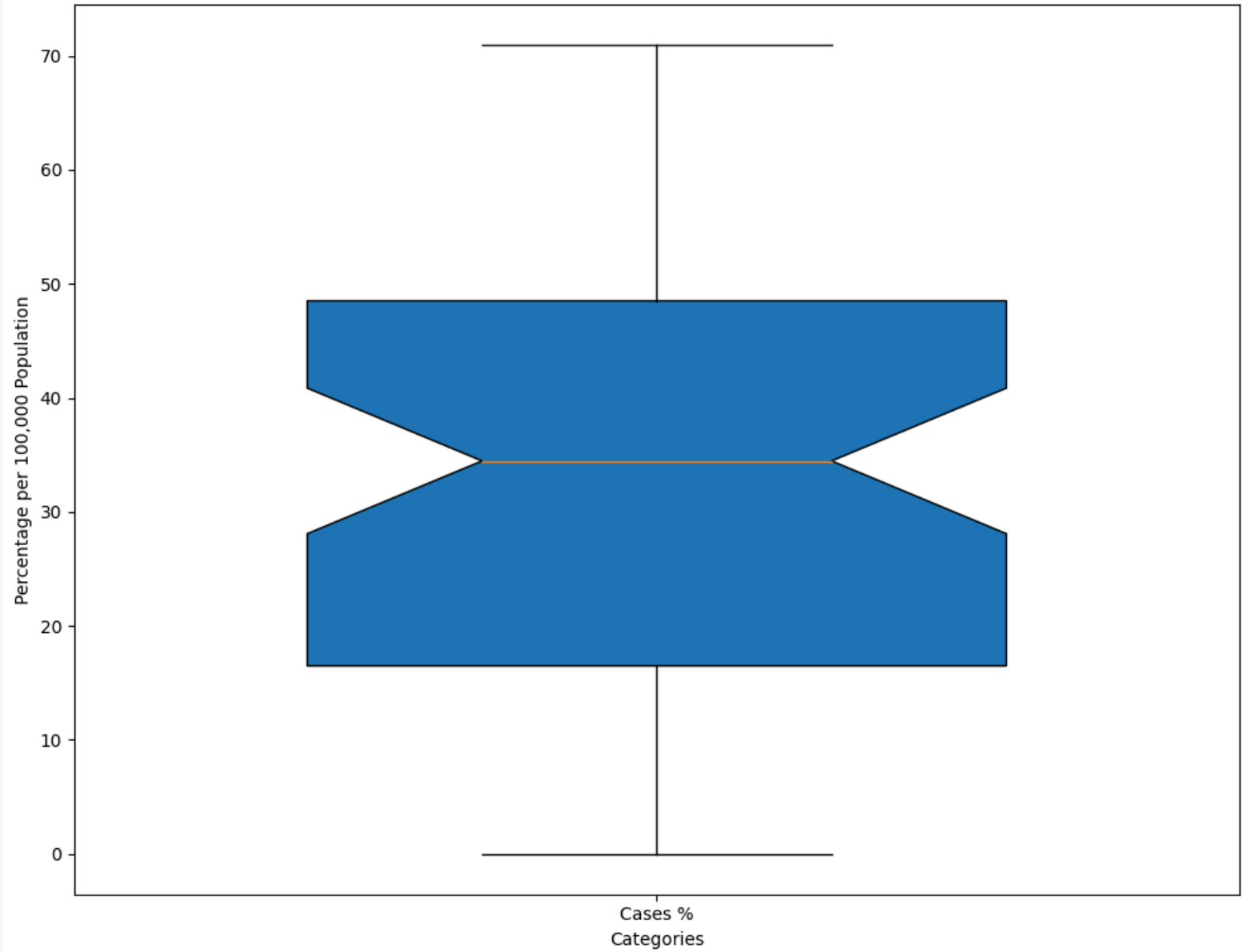
# Set x-axis label
plt.xlabel('Categories')
```

```
# Set y-axis label
plt.ylabel('Percentage per 100,000 Population')

# Add title
plt.title('Distribution of Cases per 100,000 Population in European Countries')

# Show plot
plt.tight_layout()
plt.show()
```

Distribution of Cases per 100,000 Population in European Countries



The visualisation provides a comparative analysis of COVID-19 statistics across various European countries. It focuses on the percentage of cumulative cases and per 100,000 population in Europe. The box plot helps the user to visualize a concise summary of the distribution of a dataset, displaying key statistics like the median, quartiles, and potential outliers.

This visualisation allows for a quick and intuitive assessment of the impact of COVID-19 in European nations. It vividly highlights disparities in case rates, making it easy to identify countries with higher or lower percentages in relation to their population. The information is obtained from the WHO-COVID-19-global-table-data" dataset, with the Python code specifying European Countries only.

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Visualisation 5:

What is the comparison between percentage of Covid Cases and number of vaccine types used in European Countries?

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

# Load data from the first CSV file (COVID-19 data)
df_covid_data = pd.read_csv('WHO-COVID-19-global-table-data.csv')

# Filter data for European countries
df_covid_europe = df_covid_data[df_covid_data['WHO Region'] == 'Europe']

# Calculate percentage of cases per 100,000 population
df_covid_europe['Cases_percentage'] = (df_covid_europe['Cases - cumulative total per 100000
population'] / 100000) * 100

# Load data from the second CSV file (Vaccination data)
df_vaccination_data = pd.read_csv('vaccination-data.csv')

# Merge DataFrames based on a common identifier (e.g., 'Country')
merged_df = pd.merge(df_covid_europe, df_vaccination_data, on='Country', how='inner')

plt.figure(figsize=(12, 8)) # Increase figure size for better spacing

plt.plot(merged_df['Country'], merged_df['Cases_percentage'], marker='o', label='Cases Percentage',
color='blue')
plt.plot(merged_df['Country'], merged_df['NUMBER_VACCINES_TYPES_USED'], marker='o', label='Number of
Vaccine Types Used', color='green')

# Add a line of best fit for blue data points
coefficients_blue = np.polyfit(range(len(merged_df)), merged_df['Cases_percentage'], 1)
line_of_best_fit_blue = np.polyval(coefficients_blue, range(len(merged_df)))
```

```

plt.plot(merged_df['Country'], line_of_best_fit_blue, color='blue', linestyle='--', label='Line of
Best Fit (Cases)')

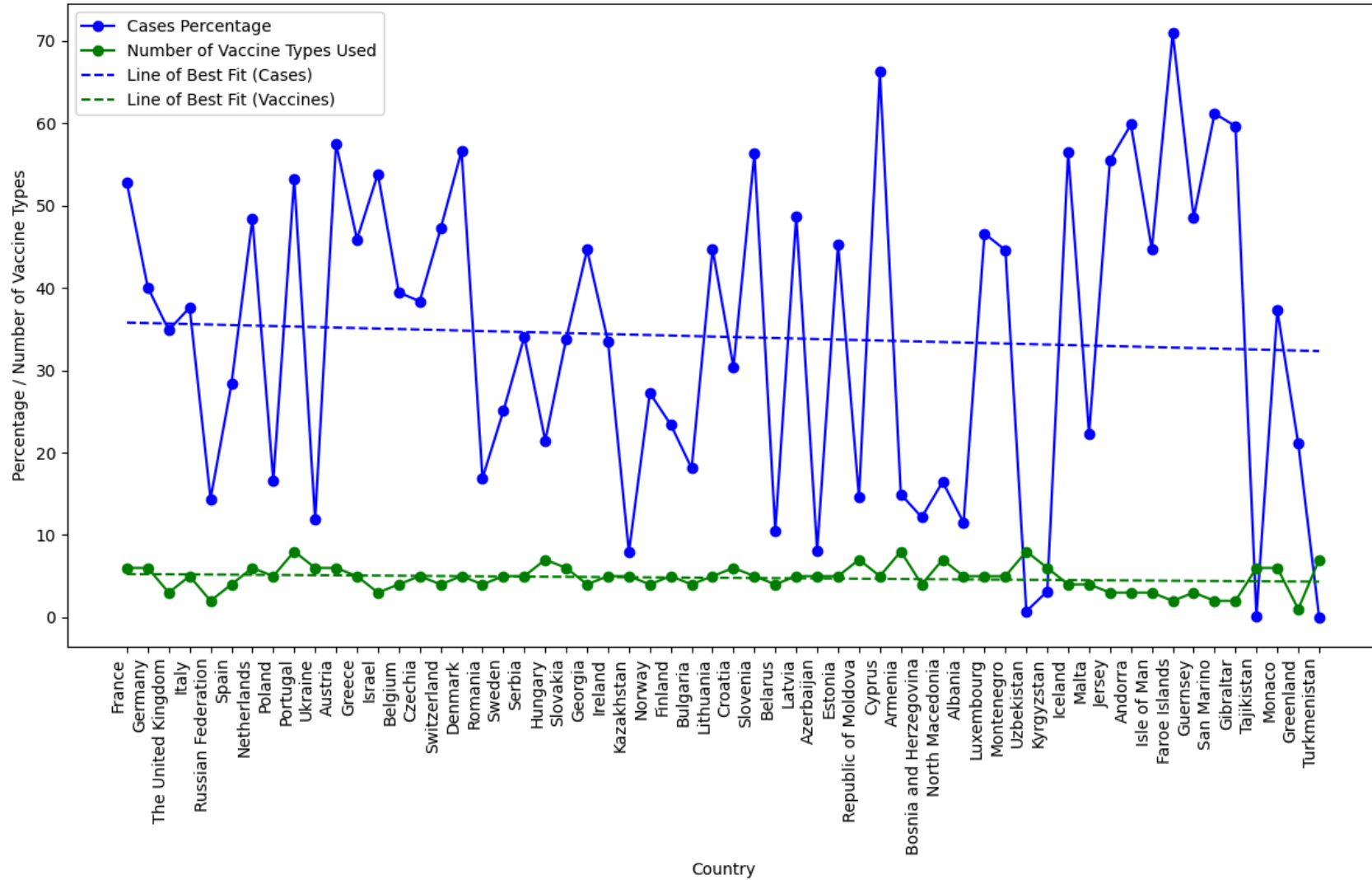
# Add a line of best fit for green data points
coefficients_green = np.polyfit(range(len(merged_df)), merged_df['NUMBER_VACCINES_TYPES_USED'], 1)
line_of_best_fit_green = np.polyval(coefficients_green, range(len(merged_df)))
plt.plot(merged_df['Country'], line_of_best_fit_green, color='green', linestyle='--', label='Line of
Best Fit (Vaccines)')

plt.xlabel('Country')
plt.ylabel('Percentage / Number of Vaccine Types')
plt.title('Comparison of Covid Cases Percentage and Number of Vaccine Types Used in European
Countries')
plt.xticks(rotation=90, ha='right') # Rotate and align x-axis labels for better readability
plt.tight_layout() # Adjust layout to prevent labels from being cut off
plt.legend()

plt.show()

```


Comparison of Covid Cases Percentage and Number of Vaccine Types Used in European Countries



This visualisation compares the percentage of COVID-19 cases per 100,000 population with the number of vaccine types used in various European countries. Leveraging data from two distinct CSV files (WHO-COVID-19-global-table-data.csv and vaccination-data.csv) the blue line illustrates the percentage of cases, while the green line represents the number of vaccine types administered.

The x-axis showcases the European countries, with their names rotated vertically for enhanced legibility. Each point on the graph corresponds to a specific country, providing a visual representation of how the percentage of cases and the variety of vaccine types vary across the region. Dashed lines of best fit for both cases and vaccine types offer further insights, potentially revealing underlying trends or correlations in the data.

Visualisation 6

What are the percentage of total cumulative total cases for: UK, Russia and USA?

```
##ADDITIONAL VISUALISATION###

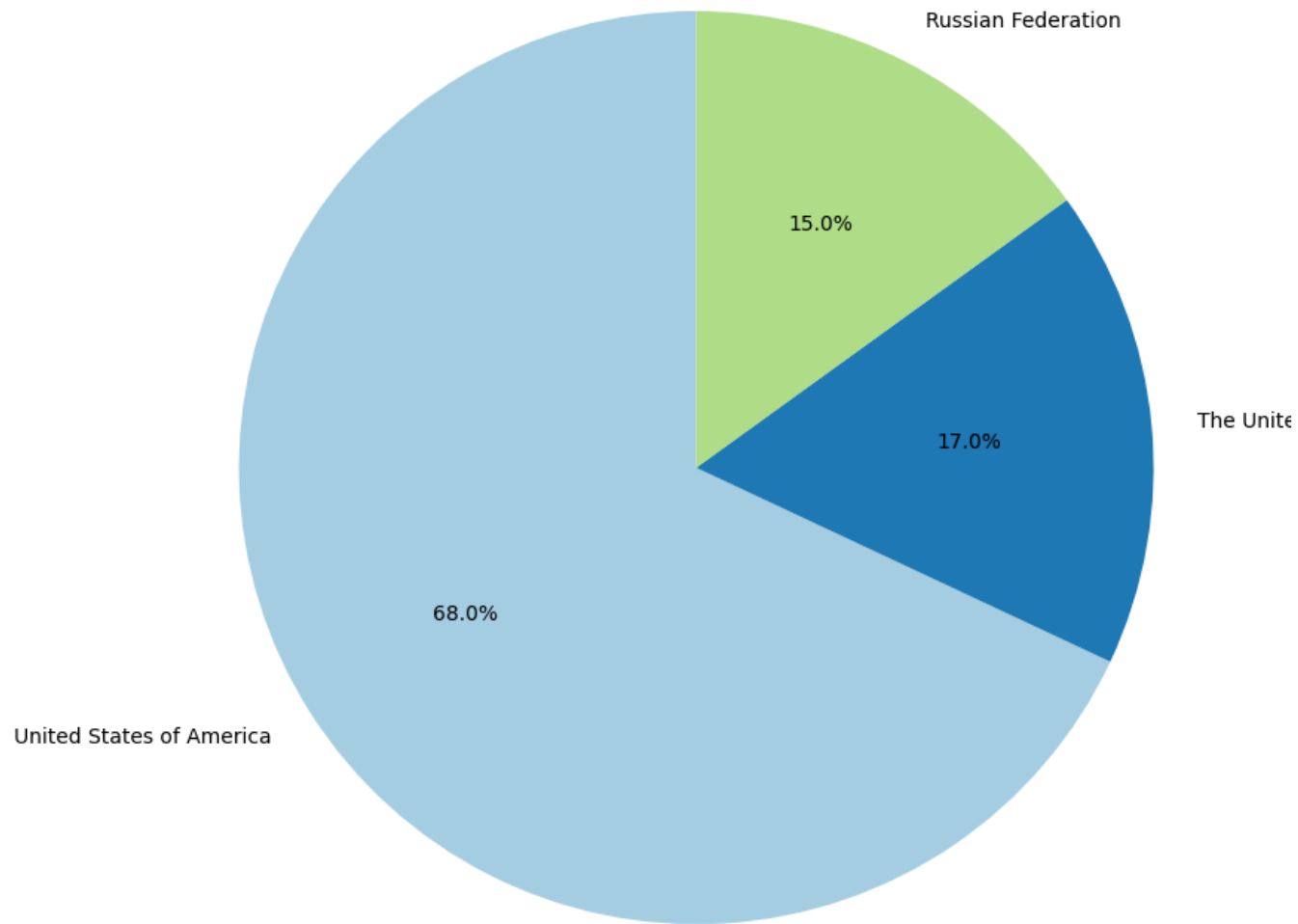
import pandas as pd
import matplotlib.pyplot as plt

# Load the data from the CSV file
file_path = 'WHO-COVID-19-global-table-data.csv'
df = pd.read_csv(file_path)

# Filter data for The United Kingdom, Russia, and USA
selected_countries = ['The United Kingdom', 'Russian Federation', 'United States of America']
filtered_df = df[df['Country'].isin(selected_countries)]

# Create a pie chart
plt.figure(figsize=(10, 10))
plt.pie(filtered_df['Cases - cumulative total'], labels=filtered_df['Country'], autopct='%1.1f%%',
startangle=90, colors=plt.cm.Paired.colors)
plt.title('Cumulative Total Cases for Selected Countries')
plt.show()
```

Cumulative Total Cases for Selected Countries



The provided pie chart visualises the cumulative total COVID-19 cases for The United Kingdom, Russia, and the United States of America. Each segment of the pie corresponds to one of these countries, with the size of each segment representing the proportion of total cases attributed to that specific country. The use of distinct colors enhances the differentiation between the countries, aiding viewers in easily identifying and comparing the contributions of each nation to the global cumulative case count.

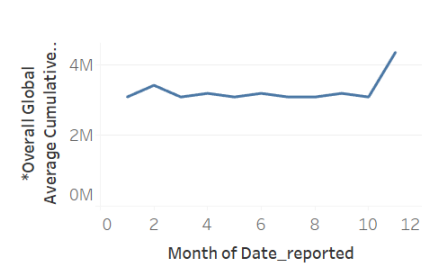
This pie chart is effective in conveying a high-level overview of the distribution of COVID-19 cases among the selected countries. It allows for a quick and intuitive comparison of the magnitude of cases for each country within the context of the global dataset. While pie charts are particularly useful for displaying the relative proportions of different components in a whole, it is important to note that they may not be suitable for displaying precise values. In this case, the emphasis is on providing a visual snapshot of the contribution of each country to the overall cumulative total, facilitating a rapid understanding of the global distribution of COVID-19 cases.

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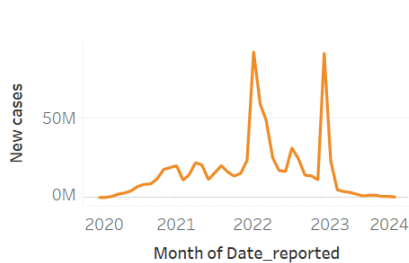
TABLEAU VISUALISATION

Dashboard:

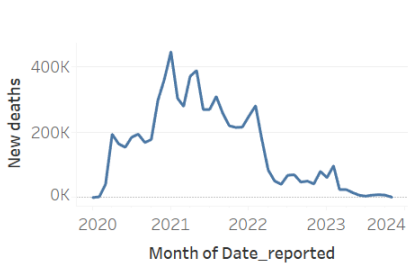
Overall Global Average
Cumulative Cases per month



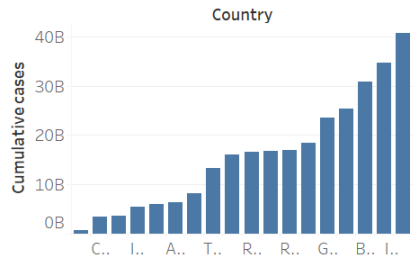
Sum of daily covid cases for
each month in the world



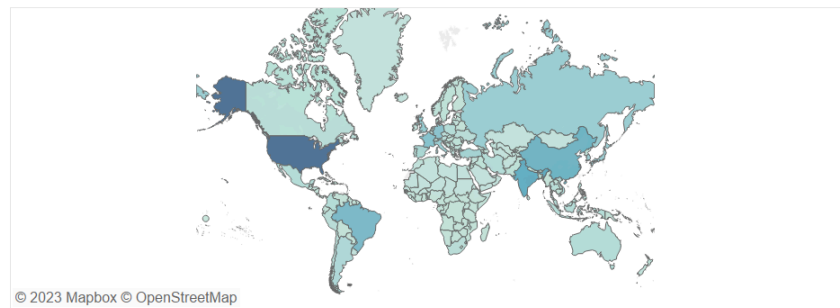
Sum of new daily deaths for
each month for the world



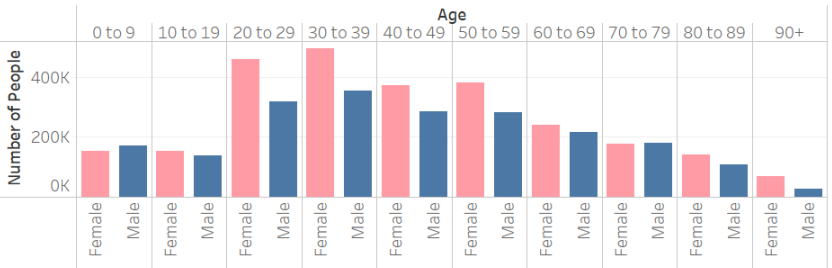
Cumulative cases for G20
Countries



Cumulative Covid Cases for all World Countries



Number of people newly tested for COVID-19, by age and gender,
28 May to 26 August 2020, England



REFERENCES

Data Source for additional graph:

<https://www.gov.uk/government/publications/demographic-data-for-coronavirus-testing-england-28-may-to-26-august/demographic-data-for-coronavirus-covid-19-testing-england-28-may-to-26-august#table-3>